

PLANRS ACADEMY

K-12 HARDWARE & PHYSICAL COMPUTING PATHWAY

LED Dice Adventure Random Number Project

Detailed Instructional Lesson Plan & Curriculum Guide

Target Group: Grade 1 (Ages 6-7)

Session Duration: 30 Minutes

Core Topic: Simple Electrical Loop Circuits, Inputs, Outputs & Randomization

Document Scope: Comprehensive 12-Page Lesson Architecture

1. Session Context & Hook Narrative

The Story of the Missing Dice

In Grade 1 curriculum design, launching a lesson with analytical definitions of electrical friction or current flow fails to establish internal student motivation. At Planrs Academy, every session begins with a relatable, real-world conflict scenario that kids naturally solve through hands-on technical construction.

This session leverages a scenario every Indian childhood household has experienced: a family rainy day board game session brought to a complete halt by a misplaced toy token component.

*"Imagine you are sitting on the floor mat on a rainy Sunday evening with all your favorite cousins, your older brother, and your grandparents. The colorful board is laid out cleanly in the middle of the circle. You have chosen the blue token, and you are just three blocks away from entering home to win the grand game of **Ludo**!"*

You shake your hands eagerly, reach out to pass the dice to your friend, and—oh no! The tiny white plastic dice slips from your finger, tumbles off the bed sheets, and rolls deep underneath the heavy wooden storage diwan. It is too dark down there to find it, and nobody can crawl underneath.

*Does the game have to stop forever? Are we stuck? No! We are smart engineers. Today, we are going to build our very own magical, glowing, electronic **LED Dice** from scratch!"*

2. Core Concept: What is a Random Number?

Demystifying Probabilities for 6-Year-Olds

Randomization is an analytical branch of mathematics and software architecture, yet for early childhood education, it must be framed through the lenses of wonder, secrecy, and unpredictability. Students learn that a random number is a delightful digital surprise—something that the computer chooses independently, preventing any player from guessing the final outcome before it actually arrives.

The Chit-Picking Folk Metaphor

To clarify how an integrated hardware chip cycles through unseen states to generate random points, the teacher links the mechanism to a traditional folk game played across schools and communities in India:

The Traditional Indian "Chit-Picking" Metaphor:

Think about what happens when you play a local game of chits at school. You take a clean piece of white paper, tear it into five small square pieces, and write down a unique friend's name or a secret score on each one. Then, you roll and fold every little paper slip up into a tight, identical ball so they all look exactly the same from the outside.

You place all the paper slips into Grandfather's old cotton topi or a kitchen steel bowl, mix them up round and round, close your eyes tightly, and pull one out. Until your fingers unfold that paper slip, the number inside remains a total mystery! That surprise factor is exactly how a computer or our electronic dice selects a random number.

3. Structural Circuit Anatomy: Meet the Parts

Introducing Components via Active Roles

Instead of presenting formal textbook definitions of positive anodes, voltage gradients, or non-conductive barriers, this lesson treats physical hardware pieces as lively characters inside a miniature workspace community. Each component plays an essential role in delivering the energy required to make the dice sparkle.

THE POWER GENERATOR

The Coin-Cell Battery

Think of the small, round silver battery as a packed water tank sitting high up on a roof terrace. It holds all the electrical juice and power ready to rush out down the pipes whenever a channel is cleared for it.

THE GATEKEEPER

The Push-Button Switch

The spring-loaded button acts like a physical drawing room door or a garden water tap. When it sits untouched, the door remains closed and locked tight, blocking energy flow. When small fingers press down on it, the door swings wide open, creating a clear pathway for power.

THE VISUAL REWARD

The LED Light Matrix

The Light Emitting Diode is the final destination. It receives the running current and instantly transforms it into a brilliant, glowing dot pattern, showing us our secret rolled number.

4. Systems Logic: Input vs. Output Relationships

Mapping Action to Reaction

An essential computational thinking competency for Grade 1 is recognizing cause-and-effect loops in automated machines. Children learn that technology functions by pairing human-driven actions (**Inputs**) with machine-driven responses (**Outputs**).

The Household Doorbell Analog:

*"Think about walking up to your neighbor's flat or your aunt's house. You stretch out your thumb and press down firmly on that small plastic button mounted on the door wall. That push is your **Input**. Instantly, from deep inside the kitchen hallway, an electric chime rings out 'Ding-Dong!' The sound you hear is the house's **Output**!"*

Without your finger's input, the chime remains silent. In our electronic dice, pushing our button is the Input, and the sudden flash of glowing dots on our matrix is the beautiful, brilliant Output!"

Your Classroom Input Action	The Corresponding Machine Output Result
You press down on the dice's push button.	The circuit links together, sending power into the microchip matrix.
The internal processor picks a surprise chit.	Specific LED lights ignite, displaying a cluster of dots.
You release your finger away from the button.	The path opens, current drops, and the glowing dots rest safely.

5. Electrical Architecture: The Complete Circuit Loop

The Concept of Unbroken Continuity

Electricity cannot leap across empty air spaces; it behaves like a flowing stream that demands an absolute, uninterrupted loop to proceed. If any point along the metal line suffers a break, slice, or loose junction, the entire electrical flow instantly stops, leaving the system dark.

The Temple Pradakshina Metaphor

To clarify this continuous movement to young learners in India, we frame the circuit loop as a structural path of respect and devotion found in daily community outings:

The Concept of Pradakshina / Parikrama:

When you visit a historic temple or a community celebration area with your parents, you walk along the stone path circling all the way around the central structure. This beautiful walk is called a **Pradakshina** or **Circumambulation**.

To complete your walk, you must start at the entry gate, pace smoothly all the way around the back wall, and arrive cleanly back at the front steps without skipping any stones. If a massive metal barricade blocks your path halfway through, your circle breaks, and you cannot proceed! Electricity travels the exact same way—it needs a complete temple pathway to run its course.

Translating the Parikrama to the LED Dice Board

During physical assembly, instructors point out that the wire traveling from the battery's positive face must route through the switch terminal, enter the light frame, and cycle smoothly back down into the battery's negative bottom layer to satisfy this path requirement.

6. Session Pacing Blueprint (30 Minutes Total)

Maintaining instructional velocity ensures that Grade 1 cohorts finish hardware construction before mental fatigue or distraction disrupts the learning flow. Product development teams must engineer physical kits to fit seamlessly into this pacing timeline:

Time Block	Module Identifier	Detailed Classroom Activity Breakdown & Delivery Directives
00:00 - 05:00	The Hook & Setup	Deliver the story of the missing dice beneath the wooden storage diwan. Ask students to share local board games they play at home (Ludo, Snakes & Ladders, Paramapadham).
05:00 - 12:00	Hardware Anatomy	Unpack the individual kit components. Map the battery, button, and LED blocks to real-world equivalents. Introduce the Chit-Picking game metaphor for random states.
12:00 - 25:00	Guided Kit Build	Step-by-step physical code assembly. Connect power lines, anchor terminal blocks into the housing shell, and test connection lines.
25:00 - 30:00	Ludo Validation Race	Students use their newly engineered glowing LED dice to play a rapid 5-step validation board race with bench partners to ensure system durability.

7. Step-by-Step Build Guide: Step 1 (Power Configuration)

Initializing Assembly Without Adult Anxiety

Traditional electrical engineering often presents major barriers for young elementary classrooms due to risks involving sharp tools, stripping scissors, or hot irons. The Planrs Academy hardware pathway bypasses these challenges by utilizing pre-insulated plug-and-click magnetic connection ribbons.

Step 1 Classroom Delivery Directives

Instructors will guide the class to hold the round silver battery cell up between their fingertips, inspecting its faces before making any terminal link updates.

Step 1 Action Checklist:

- Locate the flat, smooth face of the coin-cell battery marked with a small plus symbol (+).
- Pick up the red power lead connector clip from your kit compartment.
- Slide the metallic terminal mouth of the clip onto the edge of the silver cell until a solid 'Click' is felt.

Pedagogical Reframing

"Look at that, young engineers! By connecting our lead, we have successfully opened up the main gate of our electrical water tank. The energy is now safe and flowing down our red wire, waiting patiently for our next structural command!"

8. Step-by-Step Build Guide: Step 2 (The Gatekeeper Terminal)

Integrating the Controller Interrupter

If we connect our power lines straight into our light sources without adding an intermediary switch, the lights will stay turned on constantly. The battery would empty its energy within minutes, and our board game would lose all its fun since there would be no distinct rolling action! Step 2 introduces the system gatekeeper.

Step 2 Classroom Delivery Directives

Students focus on anchoring the terminal paths to the hardware push-button module pins.

Step 2 Action Checklist:

- Locate the tiny square box component featuring the bright round button cap.
- Take the loose metal pin coming from the end of your positive red wire line.
- Gently push the metal ribbon terminal straight into the receiving port on the left wall of your switch block. No scissors, tape, or glue are needed!

Reinforcing Language Usage

Instructors must ensure that children use child-friendly terminology during execution: "We have just added a gatekeeper to our circuit! The power is running down the red highway, but it is now hitting a locked gate at our switch button, waiting for our physical thumb press to let it through."

9. Step-by-Step Build Guide: Step 3 (Closing the Loop)

Completing the Pradakshina Path

The final step connects our gatekeeper module to the display screen, routing energy back into the power cell to establish our parikrama loop line. This step anchors our display system cleanly inside the primary protective housing frame.

Step 3 Classroom Delivery Directives

Learners connect the final terminals to light up the display screen during active test frames.

Step 3 Action Checklist:

- Take the blue wire terminal coming out from the right side of your push-button gatekeeper.
- Plug that blue terminal directly into the base socket of your central LED matrix block.
- Take the final black return wire coming from the bottom of the LED block, and clip it securely to the negative (-) face of your battery.
- Press the assembly down into the pre-molded slots inside your colorful plastic housing shell.

The Grand Verification Test

"Now, let's test our work! Press down firmly on the button with your thumb. Wow! Look at that magic glow! The bright light dots are cycling rapidly and showing a surprise random number every single time you press down! Your temple circle is complete, and your custom electronic dice is ready for game time!"

10. Things to Avoid: Critical Safety & Design Constraints

Early childhood engineering design environments demand meticulous safety filters to ensure physical and emotional security. Content developers and lab curators must strictly abide by these four core constraints:

SAFETY MANDATE 1

Zero Sharp Tools & Solder: Content storyboards and kits must never require scissors, knives, or hot soldering irons. Grade 1 students are still developing basic fine motor control. All connections must rely on pre-stripped insulation sleeves and click-fit magnetic leads.

SAFETY MANDATE 2

No Mains AC Power Exposure: Children must never work near wall outlets or large transformer bricks. Systems are strictly limited to enclosed 3V DC coin batteries. This setup provides zero risk of mild irritation or shock, ensuring complete physical safety.

DESIGN RULE 3

Avoid Abstract Western Examples: Do not introduce western analogies like electronic roulette wheels or casino dice patterns. Keep examples focused on traditional regional spaces like local board game sets or school courtyard activities.

DESIGN RULE 4

No Critical Failure Language: Ban phrases like 'Circuit Failure', 'Short Circuit Defect', or 'Assembly Error'. If a light stays dark, the educator will reframe the issue as a sleeping firefly: 'Oh look, our Jugnu is resting! Let's check our circuit path to awaken it!'

11. Assessment Metrics & Success Criteria

Measuring Pedagogical Success

To verify lesson efficacy across all Planrs Academy learning environments, the instructional model tracks three specific target operational matrices:

SUCCESS TARGET A: ASSEMBLY COMPLETION

Target Threshold: 100% Student Success

Every participating learner must leave the session with a fully assembled, enclosed, and working LED Dice component that reliably cycles and throws distinct numbers upon active manual thumb button compression.

SUCCESS TARGET B: CONCEPT RETENTION

Target Threshold: 90% Accuracy on Input/Output Mapping

During the group wrap-up, students must demonstrate at least 90% alignment when matching physical hardware parts to system roles, successfully identifying the battery as the energy tank, the button as the gate, and the LED as the glowing output.

SUCCESS TARGET C: COMMUNITY EMPOWERMENT INDEX

Target Threshold: Active Group Sharing

At least 95% of children must express pride and emotional fulfillment by demonstrating their working hardware to a bench partner or an instructor during the final game module, consolidating their identity as confident junior tech creators.